Problem K. Parentheses and Swapping

Input file:	standard input
Output file:	standard output
Time limit:	1 second
Memory limit:	1024 megabytes

A parentheses sequence is a string composed of the characters (and). A parentheses sequence $s_{1,2,\ldots,m}$ is defined as valid if it satisfies one of the following conditions:

- s is empty.
- $s_1 =$ (, $s_m =$), and $s_{2,3,\ldots,m-1}$ is valid.
- There exists a position $k(1 \le k < n)$ such that $s_{1,2,\ldots,k}$ is valid, and $s_{k+1,k+2,\ldots,m}$ is valid.

Two positions $i, j(1 \le i < j \le m)$ in the parentheses sequence $s_{1,2,\ldots,m}$ are said to be matched if and only if both $s_{i,i+1,\ldots,j}$ and $s_{i+1,i+2,\ldots,j-1}$ are valid.

Given an integer sequence of length n, $a_1, a_2, \ldots, a_n$, where n is guaranteed to be even. We construct a new integer sequence $c_1, c_2, \ldots, c_n$ through a valid parentheses sequence $b_1, b_2, \ldots, b_n$, such that $\forall 1 \le i \le n, c_i = a_{p_i}$, where p_i is the position matched with position i in $b_{1,2,\ldots,n}$.

Find a valid parentheses sequence $b_{1,2,\ldots,n}$ such that the lexicographical order of $c_{1,2,\ldots,n}$ is minimized. If there are multiple solutions, output any one of them.

Input

The first line contains an integer $T(1 \le T \le 2.5 \times 10^5)$, indicating the number of test cases.

For each test case:

The first line contains an integer $n(1 \le n \le 5 \times 10^5)$, indicating the length of the sequence a, and it is guaranteed that n is even.

The second line contains n space-separated integers, representing $a_1, a_2, \ldots, a_n(1 \le a_i \le n)$.

It is guaranteed that the sum of n does not exceed 5×10^5.

Output

For each test case, output a line containing a string of length n, representing $b_{1,2,\ldots,n}$.

Example

standard input	standard output
5 6 4 1 5 4 1 1 4 1 2 3 2 4 1 3 1 2 2 2 1 8 8 5 2 6 1 4 3 7	((())) ()() (()) () ((()))()